

Fair, Non-Progressive Influence Control Under Backfire and Parameter Uncertainty

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Abstract

The classical influence maximization guarantee of Kempe, Kleinberg, and Tardos rests on four assumptions: progressive diffusion, a monotone submodular objective, known influence parameters, and no fairness requirement. Real deployments routinely violate all four at once: influence can be actively reversed (backfire, or negative word-of-mouth), edge probabilities are never known exactly, and interventions must not neglect a disadvantaged subpopulation. We study this combined relaxation. A sandwich approximation built from two monotone-submodular surrogates yields a $\rho(1 - 1/e)$ guarantee for *every* backfire intensity $\beta \in [0, 1)$, tight at $\beta = 0$, with a per-instance, directly computable certificate; $(1 - 1/e)$ -hardness survives backfire exactly. A closed-form Lagrangian index for the resulting 2-state/3-action per-node control problem replaces a bisection-plus-value-iteration inner loop with an $O(Km + n \log n)$ -per-round procedure, roughly two orders of magnitude faster in practice. We also identify and correct a population-weighting pathology in an existing isoelastic-welfare fairness objective that systematically under-protects small groups, replacing it with an egalitarian reweighting. These pieces combine into MF-BWI-Fair, a control policy that further adds mean-field belief propagation and Bayesian parameter tracking. On a synthetic benchmark, MF-BWI-Fair sustains spread under backfire and recovery where one-shot baselines – including independently reimplemented published algorithms (IMM, He–Kempe’s robust Saturate Greedy) – degrade sharply, and the fairness fix raises worst-off-group reach at negligible spread cost. The finding generalizes across four further synthetic graphs and three real Facebook ego-networks at increasing scale (224–532 nodes), with one disclosed exception: on the largest, most community-structured real graph, a more expensive rollout baseline retains a small, stable edge, examined in the Discussion section. We additionally validate our IMM reimplementation directly against the original authors’ own C++ code.

Index Terms

Influence maximization, submodular optimization, restless bandits, fairness, robust optimization, social networks.

Declarations

- **Funding:** No external funding was received for this work.
- **Conflicts of interest:** The author declares no conflicts of interest.

- **Data availability:** The SNAP ego-Facebook dataset used for real-network validation is publicly available (McAuley and Leskovec, 2012).
- **Prior submission:** This manuscript is original, has not been published previously, and is not under consideration elsewhere.